

ECON 0100 | Exercise B3 | Equilibrium

I Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties are bought and sold in a market with the following demand and supply curves:

$$P = 12 - \frac{Q_d}{2} \qquad P = 2 + \frac{Q_s}{2}$$

Prices are in galleons and quantity is in pasties.

Q1 | Equilibrium

- a) What is the equilibrium quantity? _____
- b) What is the equilibrium price? _____
- c) Plot the demand curve, the supply curve, and the starred equilibrium pair.

a) $Q^* = 10$ pasties. b) $P^* = 7$ galleons.

a) At equilibrium $Q_s = Q_d = Q^*$, so set supply equal to demand and solve for quantity:

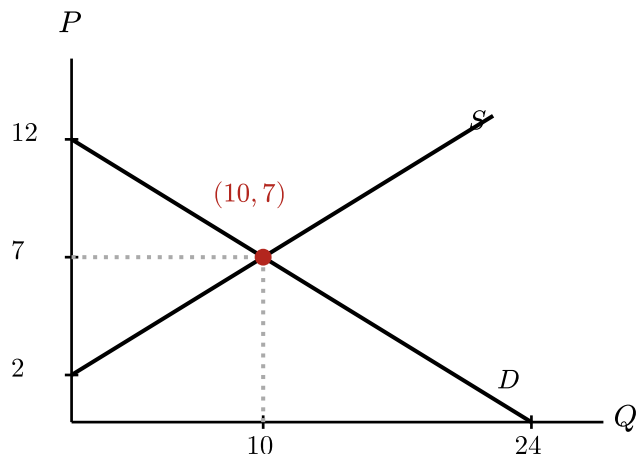
$$12 - \frac{1}{2}Q^* = 2 + \frac{1}{2}Q^* \rightarrow 10 = Q^*$$

b) Plug the quantity into either curve. Supply:

$$P^* = 2 + \frac{1}{2} \cdot 10 = 7$$

Demand gives the same: $12 - \frac{1}{2} \cdot 10 = 7$.

c) **Plot.** Demand runs from its vertical intercept 12 to its horizontal intercept 24. Supply starts at 2 and rises by 1 galleon for every 2 pasties. The star goes at $(10, 7)$, where the two curves cross: quantity first, then price.



Q2 | A Price Away from Equilibrium

Pumpkin pasties again. Suppose the price is 5 galleons.

- a) What is the quantity demanded? _____
- b) What is the quantity supplied? _____
- c) Is this a shortage or an excess, and how large? _____
- d) Which way will the price move? _____

a) 14 pasties. b) 6 pasties. c) A shortage of 8 pasties. d) Up.

a) Price to quantity along demand:

$$5 = 12 - \frac{1}{2}Q_d \rightarrow Q_d = 14$$

b) And along supply:

$$5 = 2 + \frac{1}{2}Q_s \rightarrow Q_s = 6$$

c) Below the equilibrium price buyers want more than sellers bring: $Q_d > Q_s$, a shortage of $14 - 6 = 8$ pasties, the gap between the two curves at 5 galleons.

d) The price rises. Buyers who can't get pasties offer more, and sellers with a line of willing buyers ask more, until the price reaches 7 and the gap closes. Only 6 pasties change hands while the price sits at 5: the short side of the market sets the quantity.

